

1. What will be the output of the following C application?

```
#include <stdio.h>

void main()
{
    char v[] = { 97, 99, 102 };
    char* pa = v;

    *(pa + 1) = *(pa + 1) - 1;
    printf("\na = %d, *pa = %X", v[1], pa[2]);
}
```

- a) a = 98, *pa = 66
- b) a = 99, *pa = 98
- c) a = 98, *pa = 99
- d) a = 62, *pa = 65
- e) a = 98, *pa = 98

2. What will be the content of the file **test.bin** as hexa-decimal pairs of figures (1 pair means 1 Byte) after running the following C application (little-endian approach)?

```
#include <stdio.h>

void main()
{
    long int buffer = 0x12345678;
    FILE* fp;

    fp = fopen("test.bin", "wb");

    for (int i = 0; i < sizeof(long int); i++)
        fwrite((char*)((char*)&buffer + i), sizeof(char), 1, fp);

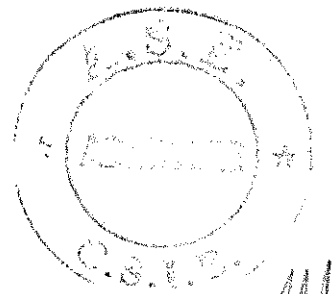
    fclose(fp);
}
```

- a) 78563412
- b) 56781234
- c) 12347856
- d) 12345678
- e) 87654321

3. What will be the output of the following C application?

```
#include <stdio.h>

char fl(char x, int* y)
{
```



```

        (*y)++;
        x += *y;

        return x;
    }

char f2(char &x, int y)
{
    y++;
    x += y;

    return x + y;
}

void main()
{
    char a = 0x65;
    int b = 1;

    a = f1(a, &b);
    b = f2(a, b);

    printf("%d %d\n", a, b);
}

```

- a) 106 109
- b) 106 110
- c) 101 109
- d) 101 110
- e) 105 109

4. What will be the output of the following C++ application?

```

#include <stdio.h>

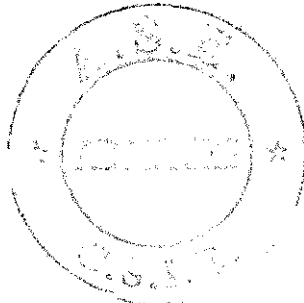
void f(char x, int *y)
{
    y++;
    x += y;
}

void main()
{
    char a = 65;
    char b = 1;

    f(b, (int*)&a);

    printf("%d %d", a, b);
}

```



- a) Compilation error triggered by the implementation of function f
- b) Compilation error triggered by the call to the function f
- c) 65 1
- d) 65 66
- e) 66 1

5. What will be the output of the following C application?

```
#include <stdio.h>
#include <string.h>

void main()
{
    char pattern[] = "exam";
    char* pv = pattern + 1;

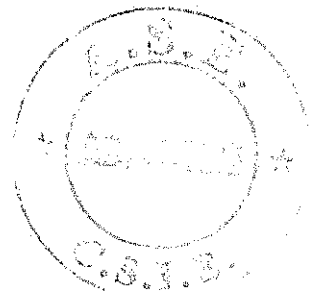
    if (pv == pattern)
        printf("Same content\n");
    else
        printf("Different content\n");

    if (&pv != (char**)pattern)
        printf("Different content\n");
    else
        printf("Same content\n");

    if (*pv == pattern[1])
        printf("Same content\n");
    else
        printf("Different content\n");

    if (!strcmp(pv, pattern))
        printf("Same content\n");
    else
        printf("Different content\n");
}
```

- a)
 - Different content
 - Different content
 - Same content
 - Different content
- b)
 - Same content
 - Same content
 - Same content
 - Same content
- c)
 - Different content



- Same content
Different content
Different content
- d)
Different content
Different content
Different content
Same content
- e)
Same content
Different content
Different content
Same content
6. What is the right choice to make a valid string size allocation and content use of the heap memory in C++ code?
- a)

```
char vstr[] = "tst";
char* str = new char[strlen("Exam")];
str[strlen(vstr)] = 0;
strncpy(str, vstr, strlen(vstr));
```
- b)

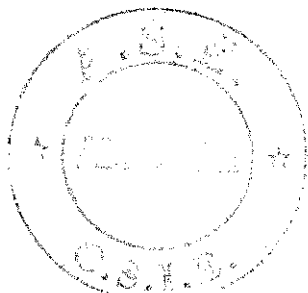
```
char vstr[] = "tst";
char* str = new char[strlen("Exam")];
strncpy(str, vstr, strlen(vstr));
```
- c)

```
char vstr[] = "tst";
char* str = new char[strlen("Exam")];
str[strlen(vstr)] = '0';
strncpy(str, vstr, strlen(vstr));
```
- d)

```
char vstr[] = "tst";
char* str = new char[strlen("Exam" + 1)];
memset(str, 0, strlen("Exam" + 1));
strncpy(str, vstr, strlen(vstr));
```
- e)

```
char vstr[] = "tst";
char* str = new char[strlen("Exam") - 1];
strncpy(str, vstr, strlen("Exam") - 1);
```
7. What will be the output of the following C application (x86 architecture C compiler, pointers are FAR by default)?

```
#include <stdio.h>
```



```
#include <string.h>

void main()
{
    char str[] = "A string";
    char* pstr = new char[strlen(str) + 1];
    strcpy(pstr, str);

    printf("%d %d %d %d\n", strlen(str), sizeof(str), sizeof(pstr), sizeof(*pstr));
}
```

- a) 8941
- b) 8841
- c) 8944
- d) 9941
- e) 9844

8. In the following C++ application, how many times the destructor of the class `Student` is called?

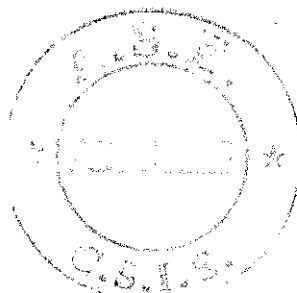
```
#include <stdio.h>

class Student
{
private:
    int age;
    char* name;
public:
    Student(int v = 0)
    {
        this->age = v;
        this->name = NULL;
    }
};

void main()
{
    Student x;
    int age1 = 21, age2 = 20;

    Student s1(age1);
    Student s2(age2);
    for (int i = 1; i <= 3; i++)
    {
        Student s3 = s1;
    }
    x = s2;
}
```

- a) 6 times
- b) 5 times
- c) 0 times



- d) 7 times
- e) There is no destructor attached to the class **Student**

9. In the following C++ application, how many times the operator = is called for the objects Students?

```
#include <stdio.h>
#include <string.h>

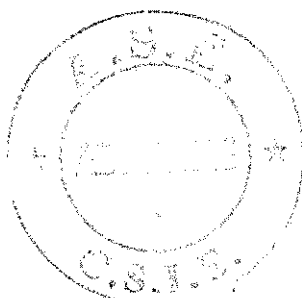
class Student
{
public:
    int age;
    char* name;
public:
    Student(int v = 0, char* sname = NULL)
    {
        this->age = v;
        if (sname != NULL)
        {
            this->name = new char[strlen(sname) + 1];
            strcpy(this->name, sname);
        }
        else
            this->name = NULL;
    }

    Student(const Student& s)
    {
        this->name = new char[strlen(s.name) + 1];
        strcpy(this->name, s.name);
        this->age = s.age;
    }

    ~Student()
    {
        if (this->name)
            delete[] this->name;
    }
};

void main()
{
    int age1 = 21;

    Student s1(age1, "John");
    Student s2 = s1;
    Student s3 = s1;
    Student s4;
    s4 = s3;
}
```



- a) 1 single time
- b) 3 times
- c) No call to overloaded operator =
- d) 4 times
- e) Compilation error because there is no overload of operator = implemented by the class **Student**

10. In C++, the default constructor is called when

- a) **An object is created by using predefined values for its attribute members**
- b) An object is created from another object of the same type
- c) An object is passed to a method by its reference
- d) An object is returned by its reference as result of a method
- e) An object is initialized by using the attribute member values of other object

11. What will be the output of the following C++ application?

```
#include <iostream>
using namespace std;

class MyClass {
public:
    static int n;

    MyClass()
    {
        n++;
    }

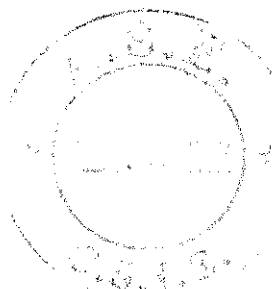
    ~MyClass()
    {
        n--;
    }
};

int MyClass::n = 0;

void main() {
    MyClass mc1, mc2, *pmc;
    pmc = new MyClass;

    cout << MyClass::n << " ";
    pmc = &mc2;
    cout << MyClass::n << endl;
}
```

- a) **3 3**
- b) 3 2
- c) 2 2
- d) 2 3
- e) 3 1



12. What will be the output of the following C++ application?

```
#include <iostream>
using namespace std;

class MyClass {
    char x;
public:
    static int n;

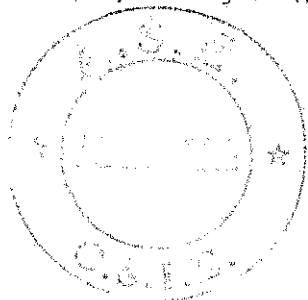
    MyClass(char in = '0')
    {
        this->x = in;
        n++;
    }
    ~MyClass()
    {
        n--;
    }
    void change(char in)
    {
        this->x += in;
    }
    char getX()
    {
        return this->x;
    }
    void setX(char in)
    {
        this->x -= in;
    }
};

int MyClass::n = 0;

void change(MyClass& obj, char in)
{
    obj.setX(in);
}

void main() {
    MyClass mc1(67), mc2(67), *pmc;
    change(mc1, 2);
    mc2.change(2);
    mc1 = mc2;
    pmc = &mc1;
    pmc->change(2);

    printf("%d %d\n", mc1.getX(), mc2.getX());
}
```



- a) 71 69
- b) 69 69
- c) 65 65
- d) 69 71
- e) 71 71

13. What will be the output of the following C++ application?

```
#include <iostream>
using namespace std;

class A
{
public:
    void print()
    {
        cout << "Class A" << endl;
    }
};

class B
{
public:
    B(const A& x) {}
    B() {}

    B& operator= (const A& x) { return *this; }

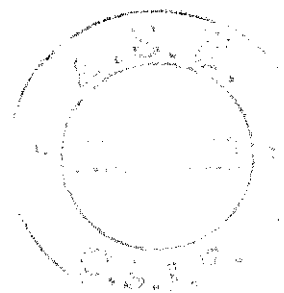
    void print()
    {
        cout << "Class B" << endl;
    }
};

void main()
{
    A ob_A;
    B ob_B1, ob_B2 = ob_A;

    ob_B1 = ob_A;
    ob_B1.print();
    ob_B2.print();

    ob_A.print();
}
```

- a) Class B
- Class B
- Class A



b) Compilation error message

c)

Class B
Class A
Class A

d)

Class A
Class A
Class A

e)

Class A
Class B
Class A

14. What will be the output of the following C++ application?

```
#include <iostream>
using namespace std;

class String
{
private:
    int length;
    char* ps;
public:
    String();
    String(const char*);
    ~String();
    const String& operator=(const String&);
    const String& operator+(const String&);
    const String& operator+(const char*);
    char* getString() { return this->ps; }
};

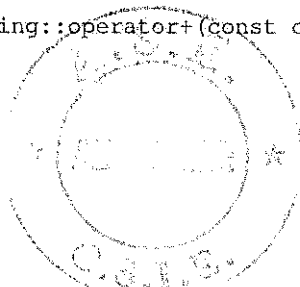
String::String() {
    this->length = 0;
    this->ps = NULL;
}

String::String(const char* str) {
    this->length = strlen(str);
    this->ps = new char[this->length + 1];
    strcpy(this->ps, str);
}

String::~String() {
    if (this->ps) delete[] this->ps;
    this->length = 0;
    this->ps = NULL;
}

const String& String::operator+(const String& strSrc) {
    String* tempS;
    tempS = new String();
    tempS->length = this->length + strSrc.length;
    if (tempS->ps != NULL) delete[] tempS->ps;
    tempS->ps = new char[tempS->length + 1];
    strcpy(tempS->ps, this->ps);
    strcat(tempS->ps, strSrc.ps);
    return (*tempS);
}

const String& String::operator+(const char* str) {
```



```

String* tempS;
tempS = new String();
tempS->length = this->length + (int)strlen(str);
if (tempS->ps != NULL) delete[] tempS->ps;
tempS->ps = new char[tempS->length + 1];
strcpy(tempS->ps, this->ps);
strcat(tempS->ps, str);
return (*tempS);
}
const String& String::operator=(const String& strSrc) {
    if (this != &strSrc) {
        this->length = strSrc.length;
        if (this->ps) delete[] this->ps;
        this->ps = new char[this->length + 1];
        strcpy(this->ps, strSrc.ps);
    }
    return *this;
}

```

```

void main()
{
    String s1("The first string ");
    String s2("and the second one. ");
    String s3("A string. ");

    s3 = s3 + "Plus a new string. ";
    s3 = s3 + s2.getString();
    cout << s3.getString() << endl;
}

```

- a) "A string. Plus a new string. and the second one."
- b) "The first string and the second one. Plus a new string. "
- c) "Plus a new string. "
- d) Runtime errors due to heap memory sharing
- e) Compilation error message due to wrong implementation of the overloaded operators

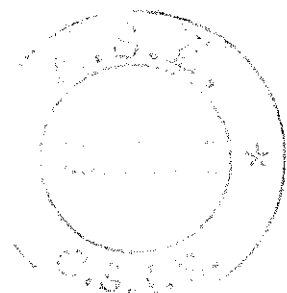
15. In the following C++ application, how many times the operator **delete** is called?

```

#include <stdio.h>
#include <string.h>

class Student
{
public:
    int age;
    char* name;
public:
    Student(int v = 0, char* sname = NULL)
    {
        this->age = v;
        if (sname != NULL)
        {
            this->name = new char[strlen(sname) + 1];
            strcpy(this->name, sname);
        }
        else
            this->name = NULL;
    }
}

```



```

    }
    Student(const Student& s)
    {
        this->name = new char[strlen(s.name) + 1];
        strcpy(this->name, s.name);
        this->age = s.age;
    }
    Student operator+(Student s)
    {
        Student tstud;
        tstud.age = (this->age + s.age) / 2;
        tstud.name = new char[strlen(this->name) + strlen(s.name) + 1];
        strcpy(tstud.name, this->name);
        strcpy(tstud.name + strlen(this->name), s.name);

        return tstud;
    }
    ~Student()
    {
        if (this->name)
            delete[] this->name;
    }
    void operator=(Student& s)
    {
        if (this->name)
            delete[] this->name;
        this->name = new char[strlen(s.name) + 1];
        strcpy(this->name, s.name);
        this->age = s.age;
    }
};

void main()
{
    int age1 = 21, age2 = 20;
    Student s1(age1, "John");
    Student s2(age2, "James");
    s2 = s1 + s2;
    Student s3 = s1 + s2;
    Student s4;
    s4 = s3;
}

```

- a) 10 times
- b) Compilation errors due do wrong call to copy constructor
- c) 9 times
- d) Compilation errors due do wrong call to operator +
- e) 11 times

1. What will display the following Java code?

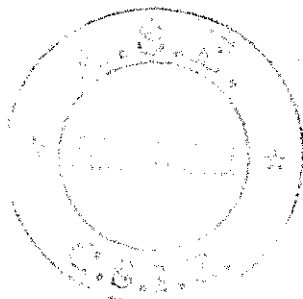
```

package eu.uni.jtest;

import java.util.HashMap;
import java.util.Map;

final class Url {
    private final String protocol, domain, uri;
    private final Integer port;

```



```

    public Url(String protocol, String domain, Integer port, String uri) {
        this.protocol = consistencyCheck(protocol, "http");
        this.domain = domain;
        this.port = Integer.decode(consistencyCheck("" + port, "80"));
        this.uri = uri;
    }

    private static String consistencyCheck(String val, String expectedVal) {
        if (val.compareTo(expectedVal) != 0)
            throw new IllegalArgumentException(val + ": " + expectedVal);

        return val;
    }

    @Override
    public boolean equals(Object o) {
        if (o == this)
            return true;
        if (!(o instanceof Url))
            return false;
        Url url = (Url) o;
        return url.protocol.equals(protocol) && url.domain.equals(domain)
            && url.uri.equals(uri) && url.port.equals(port);
    }
}

public class MainTest2 {
    public static void main(String[] args) {
        Map<Url, String> m = new HashMap<>();

        Url k = new Url("http", "www.ase.ro", 80, "test");
        m.put(k, "First Url");
        String val = m.get(k);
        System.out.println("val = " + val);
    }
}

```

- a) val = First Url
- b) val = null
- c) Nothing because of compilation error
- d) Nothing because of runtime error
- e) None of the existing answers

2. What will display the following Java code?

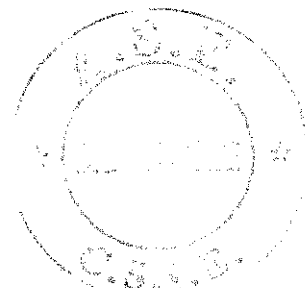
```

package eu.uni.jtest;

import java.util.Arrays;

class LinearDataStructureContainer implements Cloneable {
    private Object[] elements;
    private int size = 0;
    private static final int DEFAULT_INITIAL_CAPACITY = 5;
    public LinearDataStructureContainer() {
        this.elements = new Object[DEFAULT_INITIAL_CAPACITY];
    }
}

```



```

public void push(Object e) {
    ensureCapacity();
    elements[size++] = e;
}
public Object pop() throws Exception {
    if (size == 0)
        throw new Exception();
    Object result = elements[--size];
    elements[size] = null; // Eliminate obsolete reference
    return result;
}
private void ensureCapacity() { // Ensure space for at least one more element.
    if (elements.length == size)
        elements = Arrays.copyOf(elements, 2 * size + 1);
}
@Override protected Object clone() throws CloneNotSupportedException {
    try {
        LinearDataStructureContainer result = (LinearDataStructureContainer) super.clone();
        result.elements = elements;
        return result;
    } catch (CloneNotSupportedException e) {
        throw new AssertionError();
    }
}
@Override public String toString() {
    StringBuffer r = new StringBuffer();
    r.append("[");
    for (Object o : elements) {
        if (i > 1) r.append(", ");
        r.append(o);
        i++;
    }
    r.append("]");
    return r.toString();
}
}

public class MainTest2 {
    public static void main(String[] args)
    {
        LinearDataStructureContainer ds1 = new LinearDataStructureContainer();
        ds1.push("1"); ds1.push("5"); ds1.push("9");
        try {
            LinearDataStructureContainer ds2 = (LinearDataStructureContainer) ds1.clone();
            String s = (String)ds1.pop();
            System.out.println("ds1 = " + ds1.toString() + ", ds2 = " + ds2.toString() + ", s = " + s);
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}

```

- a) ds1 = [1, 5, null, null, null], ds2 = [1, 5, null, null, null], s = 9
- b) ds1 = [1, 5, null, null, null], ds2 = [1, 5, 9, null, null], s = 9
- c) Nothing because of compilation error
- d) ds1 = [1, 5], ds2 = [1, 5, 9], s = 9
- e) None of the existing answers

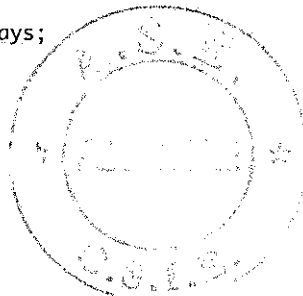
3. What will display the following Java code?

```

package eu.uni.jtest;

import java.util.Arrays;

```



```

private final Date date;

public Exam(String location, Date date) {
    this.location = location;
    this.date = (Date) date.clone();
}
public String getLocation() {
    return location;
}
public Date getDate() {
    return date;
}
@Override
public String toString() {
    return new String("location = "+location+", date = "+date);
}
}

public class MainTest2 {
    public static void main(String[] args) {
        String location = new String("Bucharest");
        Date date = new Date();
        Exam e1 = new Exam(location, date);
        Exam e2 = new Exam(location, date);
        System.out.println("p1 equals p2 - " + (e1.equals(e2)));
        System.out.println("p1 == p2 - " + (e1 == e2));
    }
}

```

- a) p1 equals p2 - false \n p1 == p2 - false
- b) p1 equals p2 - true \n p1 == p2 - false
- c) Nothing because of compilation error
- d) Nothing because of runtime error
- e) None of the existing answers

6. What will display the following Java code?

```

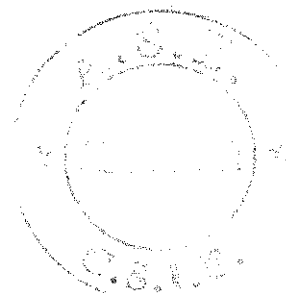
package eu.uni.jtest;

class MyGen<T> {
    T var;
    void set(T t) {
        this.var = t;
    }
    T get() {
        return var;
    }
}

public class MainTest2 {
    public static void main(String[] args) {
        var m = new MyGen<Double>();
        m.set(2021.5);
        System.out.println("" + Integer.parseInt("" + m.get()));
    }
}

```

- a) Nothing because of runtime error



```

public class MainTest2 {
    public static void main(String... args) {
        int[] collectionArray = Arrays.stream(new int[] {1, -3, 7, 9, 5})
            .filter(i -> i >= 5)
            .map(i -> i * 2).toArray();

        int sum = Arrays.stream(collectionArray).map(i -> i).sum();
        System.out.printf("sum = %d", sum);
    }
}

```

- a) sum = 42
- b) sum = 43
- c) Nothing because of compilation error
- d) Nothing because of runtime error
- e) None of the existing answers

4. What will display the following Java code?

```

package eu.uni.jtest;

interface FIPrime {
    boolean checkPrime(int num);
}

public class MainTest2 {
    public static void main(String... args) {
        FIPrime myPrimeFunc = (num) -> {
            boolean boolFlag = false;
            for (var i = 2; i <= num / 2; ) {
                if (num / i == 0) {
                    boolFlag = true;
                    break;
                }
                ++i;
            }
            return boolFlag;
        };

        int n = 301;
        System.out.printf("Is the number n = %d prime? Yes = %b", n, myPrimeFunc.checkPrime(301));
    }
}

```

- a) Is the number n = 301 prime? Yes = false
- b) Is the number n = 301 prime? Yes = true
- c) Nothing because of compilation error
- d) Nothing because of runtime error
- e) None of the existing answers

5. What will display the following Java code?

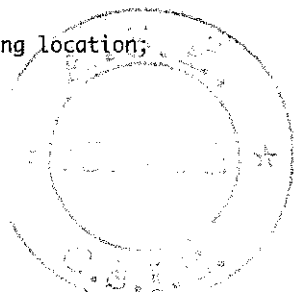
```

package eu.uni.jtest;

import java.util.Date;

class Exam {
    private final String location;
}

```



- b) 2021.5
- c) 2021
- d) Nothing because of compilation error
- e) None of the existing answers

7. What will display the following Java code?

```
package eu.uni.jtest;

import java.util.List;
import java.util.LinkedList;

public class MainTest2 {
    public static void main(String[] args) {
        List<Double> doubleLinkedList = new LinkedList<Double>();
        doubleLinkedList.add(991.5); doubleLinkedList.add(2021.0);

        for (Double d : doubleLinkedList) {
            System.out.print(d + " ");
        }
        System.out.print(" # ");
        doubleLinkedList.remove(1);
        doubleLinkedList.stream().forEach(System.out::print);

        doubleLinkedList.add(991d); doubleLinkedList.add(2022d);
        doubleLinkedList.add(1, 2018.7);

        System.out.print(" # ");
        System.out.print(" " + doubleLinkedList.get(1));
        System.out.print(" " + doubleLinkedList.indexOf(2));
        System.out.print(" " + doubleLinkedList.remove(3));
        System.out.print(" # ");

        doubleLinkedList.stream().forEach(x -> System.out.printf("%s; ", x));
    }
}
```

- a) 991.5 2021.0 # 991.5 # 2018.7 -1 2022.0 # 991.5; 2018.7; 991.0;
- b) Nothing because of compilation error
- c) 991.5 2021.0 # 2021.0 # 2018.7 -1 2022.0 # 991.5; 2018.7; 991.0;
- d) Nothing because of runtime error
- e) None of the existing answers

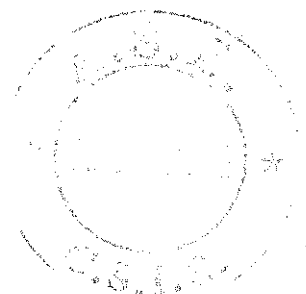
8. What will display the following Java code?

```
package eu.uni.jtest;

import java.util.List;
import java.util.ArrayList;

public class MainTest2 {
    public static void main(String[] args) {
        List<Integer> bits = new ArrayList<Integer>();
        bits.add(2); bits.add(2); bits.add(4);
        var value = 0;
        for(var i = 0; i < bits.size(); i++)
            value ^= bits.get(i);

        System.out.printf("value = %d", value);
    }
}
```



```
}  
}
```

- a) 4
- b) Nothing because of runtime error
- c) Nothing because of compilation error
- d) 5
- e) None of the existing answers

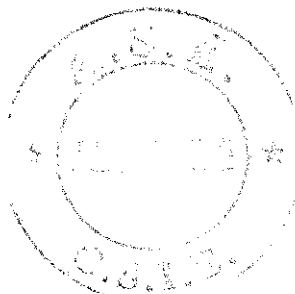
9. What will display the following Java code?

```
package eu.uni.jtest;  
  
import java.util.Arrays;  
import java.util.List;  
  
public class MainTest2 {  
    public static void main(String[] args)  
    {  
        List<Integer> numbers = Arrays.asList(100, 15, 30, 40);  
        int factor = 5;  
  
        System.out.println(  
            numbers.stream().filter(number -> number % 3 == 0)  
                .mapToInt(e -> e * factor).sum());  
    }  
}
```

- a) 225
- b) Nothing because of compilation error
- c) 280
- d) Nothing because of runtime error
- e) None of the existing answers

10. What will display the following Java code, if the given input in console is "strike" without quotes?

```
import java.io.BufferedReader;  
import java.io.IOException;  
import java.io.InputStreamReader;  
  
public class MainTest2 {  
    public static void main(String[] args)  
    {  
        String str = "";  
        BufferedReader obj = new BufferedReader(new InputStreamReader(System.in));  
        do {  
            try {  
                str = (String) obj.readLine();  
            } catch (IOException e) {  
                e.printStackTrace();  
            }  
            System.out.print(str);  
        } while(!str.equals("strike"));  
    }  
}
```



- a) **strike**
- b) Nothing because of compilation error
- c) Hello strike
- d) Nothing because of runtime error
- e) None of the existing answers

11. What will display the following Java code if run within an empty directory/folder?

```
package eu.uni.jtest;

import java.io.File;
import java.io.IOException;

public class MainTest2 {
    public static void main(String[] args)
    {
        String name = null;
        File file = null;
        try {
            file = new File("exam.txt", name);
        } catch (Exception ioe) {

        }
        if(file == null || file.exists())
            System.out.print("Exp01");
        else
            System.out.print("Exp02");
    }
}
```

- a) **Exp01**
- b) compilation error
- c) Exp02
- d) Runtime error
- e) None of the existing answers

12. What will display the following Java code?

```
package eu.uni.jtest;

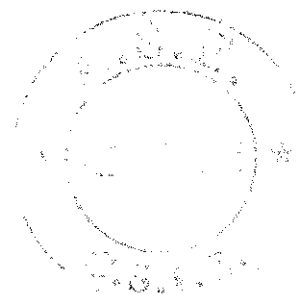
interface I {
    public default void f() {
        System.out.println("This is a default method within interface");
    }
}

abstract class A implements I {
    private int i = 207;
}

class B extends A {
    private int i = 133;

    @Override
    public void f() {
        System.out.println("i = " + this.i);
    }
}

public class MainTest2
```



```
{
    public static void main(String[] args) {
        I a = new B();
        a.f();
    }
}
```

- a) 133
- b) compilation error
- c) "This is a default method within interface" - text without quotes
- d) runtime error

13. What will display the following Java code?

```
package eu.uni.jtest;

class A {
    public A() {
        System.out.print("# Class A Constr.");
    }
}

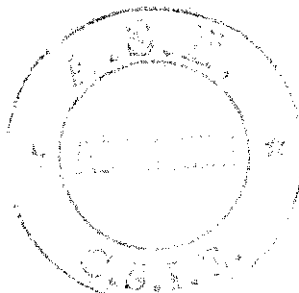
class B extends A {
    public B() {
        System.out.print("# Class B Constr.");
    }
}

class C extends B {
    private int x = 0;
    public C() {
        System.out.print("# Class C Constr.");
    }

    public C(C c) {
        System.out.print("# Class C Copy Constr.");
        this.x = c.x;
    }
}

public class MainTest2 {
    public static void main(String[] args) {
        C c1 = new C();
        C c2 = new C(c1);
        c1.toString(); c2.toString();
    }
}
```

- a) # Class A Constr. # Class B Constr. # Class C Constr. # Class A Constr. # Class B Constr. # Class C Copy Constr.
- b) compilation error
- c) # Class A Constr. # Class B Constr. # Class C Constr. # Class C Copy Constr.
- d) runtime error
- e) None of the existing answers



14. How will be displayed the "car.txt" in hex file after running the following Java code?

```
package eu.uni.jtest;

import java.io.DataOutputStream;
import java.io.FileOutputStream;
import java.io.IOException;

class Car {
    private int weight = 0;
    private String type = "";
    private String color = "";

    public Car(int weight, String type, String color){
        this.weight = weight; this.type = type; this.color = color;
    }
    public Integer getWeight() { return weight;}
    public void setWeight(Integer weight) { this.weight = weight; }
    public String getType() { return type; }
    public String getColor() { return color; }
    public void setColor(String color) { this.color = color; }
    public String toString() {
        return "Car{" + "color='" + color + "'"
            + ", type = '" + type + "', weight=" + weight + '}'';
    }
}

public class MainTest2 {
    public static void main(String[] args)
    {
        try (FileOutputStream fos = new FileOutputStream("car.txt");
            DataOutputStream out = new DataOutputStream(fos);) {

            Car c = new Car(1024, "Audi A4", "black");
            out.writeUTF("" + c.getWeight());
            out.writeUTF(c.getType());
            out.writeUTF(c.getColor());

        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```



- a. 00 04 31 30 32 34 00 07 41 75 64 69 20 41 34 00 05 62 6C 61 63 6B
- b. 00 04 31 30 32 34 00 07 41 75 64 69 20 41 34 00 05 62 6C 61 63 6B 0D 0A
- c. 00 04 31 30 32 34 00 07 41 75 64 69 20 41 34 00 05 62 6C 61 63 6B 0A
- d. Runtime error
- e. None of the existing answers

15. What will display the following Java code?

```
package eu.uni.jtest;

import java.util.HashMap;
import java.util.Iterator;
import java.util.Set;

public class MainTest2 {
    public static void main(String[] args) {
        HashMap<String, Integer> hashMapPhoneAgenda = new HashMap<String, Integer>();
        hashMapPhoneAgenda.put("John", 783500);
        hashMapPhoneAgenda.put("Andrew", 235000);
        hashMapPhoneAgenda.put("Alex", 295600);

        Set<String> keys = hashMapPhoneAgenda.keySet();
        Iterator<String> it = keys.iterator();
        String K = it.next();
        for (; it.hasNext(); K = it.next()) {
            System.out.print("; Key = " + K + ", Value = " + hashMapPhoneAgenda.get(K));
        }
    }
}
```

- a. ; Key = Alex, Value = 295600; Key = Andrew, Value = 235000
- b. ; Key = Alex, Value = 295600; Key = Andrew, Value = 235000; Key = John, Value = 783500
- c. Compilation error
- d. Runtime error
- e. None of the existing answers

